

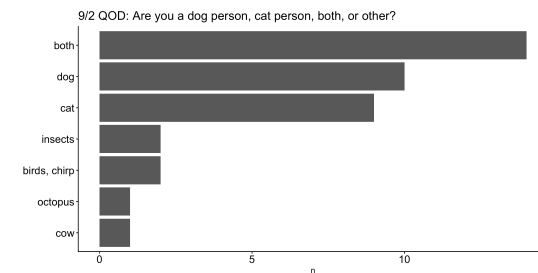


Data Visualization: the 5 Named Graphs with ggplot2

1

Reminders/Announcements

- Office hours update
- First homework assignment due on Thursday 9/10
- There won't be a quiz on this first homework



2

Last Time

- Data frames
- Motivation for data visualizations
- “Grammar” of graphics and good graphical practices

Goals for Today

- Recall our motivation for good graphics
- Learn the general structure of **ggplot2**
- Learn five standard graphs for numerical/quantitative data:
 - **Histogram**: one quantitative variable
 - **Boxplot**: one quantitative variable
 - **Barplot**: one categorical variable, one quantitative variable*
 - **Scatterplot** and **Linegraph**: two quantitative variables

3

Load Necessary Packages



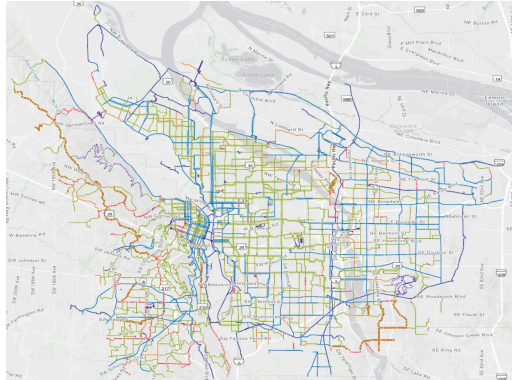
ggplot2 is part of this collection of data science packages.

```
1 # Load necessary packages
2 library(tidyverse)
```

Also, above is an example of a code comment: **# Load necessary packages**

4

Data Setting: Portland Bikeshare Data



5

Import the Data

```
1 biketown <- read.csv("data/biketown.csv")
2
3 # Inspect the data
4 glimpse(biketown)

Rows: 9,999
Columns: 19
$ RouteID      <int> 4074085, 3719219, 3789757, 3576798, 3459987, 3947695,...
$ PaymentPlan  <chr> "Subscriber", "Casual", "Casual", "Subscriber", "Casu...
$ StartHub     <chr> "SE Elliott at Division", "SW Yamhill at Director Par...
$ StartLatitude <dbl> 45.50513, 45.51898, 45.52990, 45.52389, 45.53028, 45...
$ StartLongitude <dbl> -122.6534, -122.6813, -122.6628, -122.6722, -122.6547...
$ StartDate    <chr> "8/17/2017", "7/22/2017", "7/27/2017", "7/12/2017", "...
$ StartTime    <chr> "10:44:00", "14:49:00", "14:13:00", "13:23:00", "19:3...
$ EndHub       <chr> "Blues Fest - SW Waterfront at Clay - Disabled", "SW ...
$ EndLatitude  <dbl> 45.51287, 45.52142, 45.55902, 45.53409, 45.52990, 45...
$ EndLongitude <dbl> -122.6749, -122.6726, -122.6355, -122.6949, -122.6628...
$ EndDate      <chr> "8/17/2017", "7/22/2017", "7/27/2017", "7/12/2017", "...
$ EndTime      <chr> "10:56:00", "15:00:00", "14:42:00", "13:38:00", "20:3...
```

6

Inspect the Data

```
1 # Look at first few rows
2 head(biketown)

  RouteID PaymentPlan      StartHub StartLatitude StartLongitude
1 4074085  Subscriber      SE Elliott at Division    45.50513    -122.6534
2 3719219   Casual      SW Yamhill at Director Park    45.51898    -122.6813
3 3789757   Casual      NE Holladay at MLK            45.52990    -122.6628
4 3576798  Subscriber      NW Couch at 2nd           45.52389    -122.6722
5 3459987   Casual      NE 11th at Holladay Park     45.53028    -122.6547
6 3947695   Casual      SW Moody at Thomas          45.49429    -122.6719
  StartDate StartTime                               EndHub EndLatitude
1 8/17/2017 10:44:00 Blues Fest - SW Waterfront at Clay - Disabled 45.51287
2 7/22/2017 14:49:00                               SW 2nd at Pine 45.52142
3 7/27/2017 14:13:00 NE Alberta at NE 29th/30th - Community Corral 45.55902
4 7/12/2017 13:23:00                               NW Raleigh at 21st 45.53409
5 7/3/2017 19:30:00                               NE Holladay at MLK 45.52990
6 8/8/2017 10:01:00                               SW 3rd at Ankeny 45.52248
```

Q: What does a row represent here?

7

Inspect the Data

```
1 # Determine type
2 # To access one variable: dataset$variable
3 class(biketown$BikeName)

[1] "character"

1 class(biketown$Distance_Miles)

[1] "numeric"

1 class(biketown)

[1] "data.frame"
```

8

Grammar of Graphics

- **data**: Data frame that contains the raw data
 - Variables used in the graph
- **geom**: Geometric **shape** that the data are mapped to.
 - EX: Point, line, bar, text, ...
- **aesthetic**: Visual properties of the **geom**
 - EX: X (horizontal) position, y (vertical) position, color, fill, shape
- **scale**: Controls how data are mapped to the visual values of the aesthetic.
 - EX: particular colors, log scale
- **guide**: Legend/key to help user convert visual display back to the data

9

ggplot2 example code

Guiding Principle: We will map variables from the **data** to the **aesthetic** attributes (e.g. location, size, shape, color) of **geometric** objects (e.g. points, lines, bars).

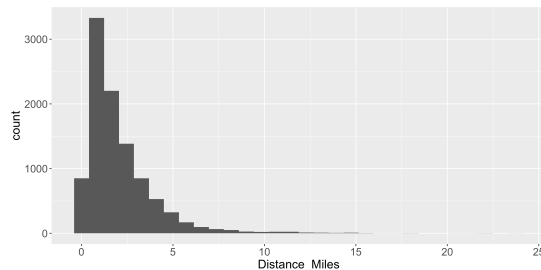
```
1 ggplot(data = ---, mapping = aes(---)) +  
2   geom_---(---)
```

- There are other layers, such as **scale_---_---()** and **labs()**, but we will wait on those for a bit.
- We are about to touch on many details of graphs - focus on recognizing this general pattern, expect to revisit these slides to refresh on the specifics

10

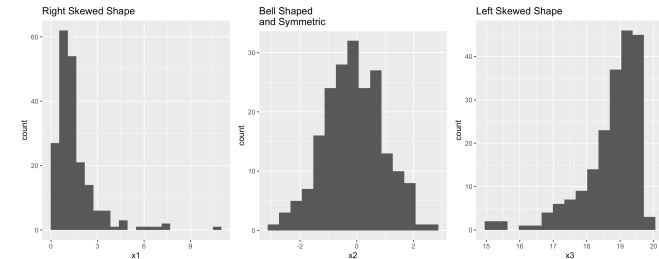
Histograms

- Binned counts of data.
- Great for assessing data distribution and shape.
- **Q**: are histograms used for *quantitative* or *categorical* variables?



11

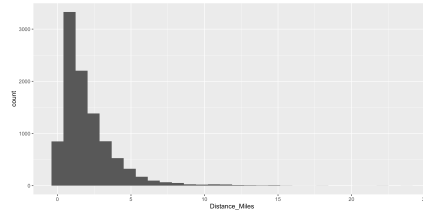
Data Shapes



12

Histograms

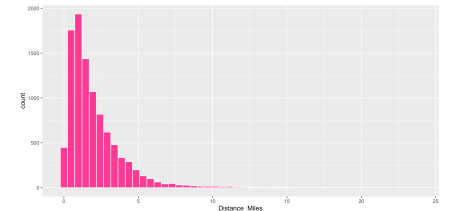
```
1 # Create histogram
2 ggplot(data = biketown,
3       mapping = aes(x = Distance_Miles)) +
4   geom_histogram()
```



13

Histograms

```
1 # Create histogram
2 ggplot(data = biketown,
3       mapping = aes(x = Distance_Miles)) +
4   geom_histogram(color = "white",
5                 fill = "violetred1",
6                 bins = 50)
```

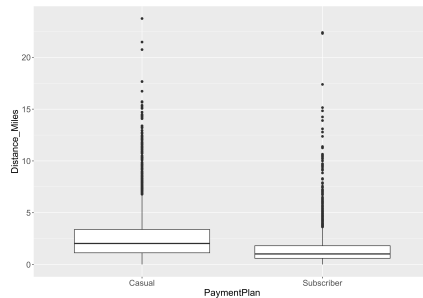


- **mapping** to a variable goes in `aes()`
- **setting** to a specific, *constant*, value goes in the `geom_---()`
- Does the right **tail** of this distribution make sense?

14

Boxplots

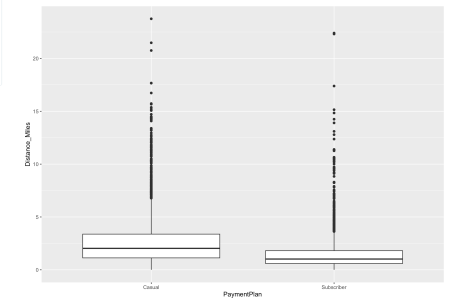
- **Five number summary:**
 - Minimum
 - First quartile (Q1)
 - Median
 - Third quartile (Q3)
 - Maximum
- Interquartile range (IQR) = $Q3 - Q1$
- Outliers: **unusual** points
 - Boxplot defines unusual as being beyond $1.5 * IQR$ from Q1 or Q3.
- Whiskers: reach out to the furthest point that is NOT an outlier



15

Boxplots

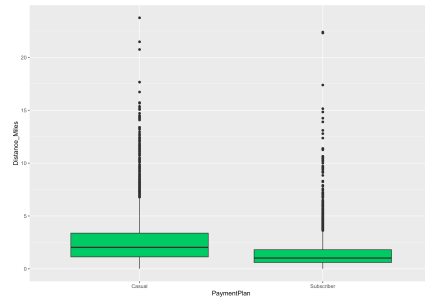
```
1 # Create boxplot
2 ggplot(data = biketown,
3       mapping = aes(x = PaymentPlan,
4                     y = Distance_Miles)) +
5   geom_boxplot()
```



16

Boxplots

```
1 ggplot(data = biketown,
2       mapping = aes(x = PaymentPlan,
3                     y = Distance_Miles)) +
4   geom_boxplot(fill = "springgreen3")
```

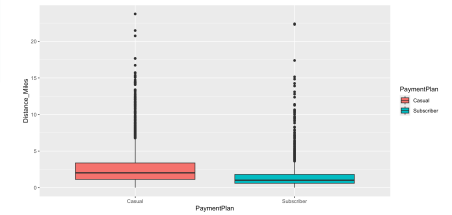


- Q: Is this **fill** an **aesthetic** mapping?

17

Boxplots

```
1 ggplot(data = biketown,
2       mapping = aes(x = PaymentPlan,
3                     y = Distance_Miles,
4                     fill = PaymentPlan)) +
5   geom_boxplot()
```

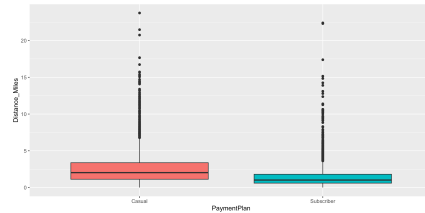


- Q: Is this **fill** an **aesthetic** mapping?
- Q: What variable is mapped to **fill**?

18

Boxplots

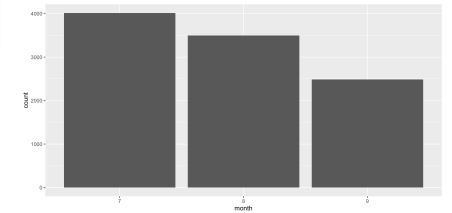
```
1 ggplot(data = biketown,
2       mapping = aes(x = PaymentPlan,
3                     y = Distance_Miles,
4                     fill = PaymentPlan)) +
5   geom_boxplot() +
6   guides(fill = "none")
```



19

Barplots

```
1 ggplot(data = biketown,
2       mapping = aes(x = month)) +
3   geom_bar()
```

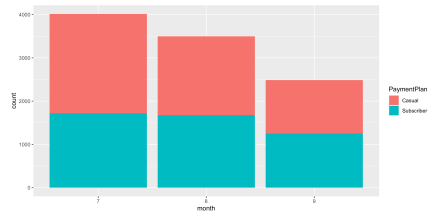


- Boxplots and histograms show the overall distribution of *quantitative* variables
- Barplots show the distribution of quantitative variables within distinct levels defined by a *categorical* variable
 - Often via **numerical summarization** of a categorical variable (counts)

20

Barplots

```
1 ggplot(data = biketown,
2       mapping = aes(x = month,
3                     fill = PaymentPlan)) +
4   geom_bar()
```

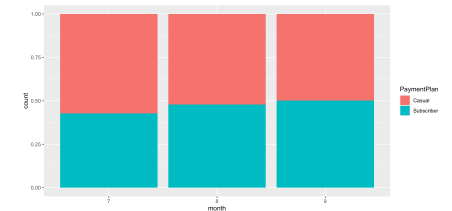


- Barplots can also show the joint distribution of *two* categorical variables via the color or fill aesthetic.
- Here, each bar is divided into separate counts with respect to the **Payment Plan** variable.

21

Barplots

```
1 ggplot(data = biketown,
2       mapping = aes(x = month,
3                     fill = PaymentPlan)) +
4   geom_bar(position = "fill")
```



- Alternatively, we can consider the y-axis to represent proportion, making direct comparison easier.
- Q:** How can we tell a histogram from a barplot?

22

New Data Context: **pdxTrees**

- The **pdxTrees** R package contains data on all the trees in the Portland Metro Area.
- Today, we'll look at the Maple, Oak, Pine, Cedar, and Douglas-fir trees in a few parks near Reed.
- Let's load the data

23

New Data Context: **pdxTrees**

- The **pdxTrees** R package contains data on all the trees in the Portland Metro Area.
- Today, we'll look at the Maple, Oak, Pine, Cedar, and Douglas-fir trees in a few parks near Reed.
- Let's load the data
- Don't worry, we haven't learned the below code yet.

```
1 library(pdxTrees)
2 near_Reed <- get_pdxTrees_parks(park = c("Woodstock Park", "Sellwood Riverfront Park", "Kenilworth Park"))
3 near_Reed <- near_Reed %>% filter(Genus %in% c("Acer", "Quercus", "Pinus", "Thuja", "Pseudotsuga"))
```

24

Inspect the data

```
1 glimpse(near_Reed)
Rows: 323
Columns: 34
$ Longitude      <dbl> -122.6304, -122.6301, -122.6301, -122.6299, ...
$ Latitude        <dbl> 45.49201, 45.49080, 45.49081, 45.49094, 45.4...
$ UserID          <chr> "7670", "7671", "7672", "7902", "7903", "79...
$ Genus           <chr> "Quercus", "Pseudotsuga", "Pseudotsuga", "Q...
$ Family          <chr> "Fagaceae", "Pinaceae", "Pinaceae", "Fagace...
$ DBH             <dbl> 3.3, 43.1, 48.2, 2.4, 11.7, 33.5, 23.5, 37...
$ Inventory_Date  <dtm> 2018-07-26, 2018-07-26, 2018-07-26, 2018-0...
$ Species         <chr> "QURU", "PSME", "PSME", "QURU", "PSME", "PS...
$ Common_Name     <chr> "Northern Red Oak", "Douglas-Fir", "Douglas...
$ Condition       <chr> "Fair", "Fair", "Fair", "Fair", "Good", "Fa...
$ Tree_Height     <dbl> 16, 148, 148, 16, 64, 118, 121, 105, 24, 12...
$ Crown_Width_NS  <dbl> 14, 61, 52, 9, 29, 32, 37, 43, 38, 44, 31, ...
```

25

Inspect the data

```
1 head(near_Reed)
# A tibble: 6 × 34
Longitude Latitude UserID Genus      Family DBH Inventory_Date Species
<dbl> <dbl> <chr> <chr> <chr> <dbl> <dtm> <chr>
1 -123. 45.5 7670 Quercus Fagac... 3.3 2018-07-26 00:00:00 QURU
2 -123. 45.5 7671 Pseudotsuga Pinac... 43.1 2018-07-26 00:00:00 PSME
3 -123. 45.5 7672 Pseudotsuga Pinac... 48.2 2018-07-26 00:00:00 PSME
4 -123. 45.5 7902 Quercus Fagac... 2.4 2018-07-26 00:00:00 QURU
5 -123. 45.5 7903 Pseudotsuga Pinac... 11.7 2018-07-26 00:00:00 PSME
6 -123. 45.5 7905 Pseudotsuga Pinac... 33.5 2018-07-26 00:00:00 PSME
# 26 more variables: Common_Name <chr>, Condition <chr>, Tree_Height <dbl>,
# Crown_Width_NS <dbl>, Crown_Width_EW <dbl>, Crown_Base_Height <dbl>,
# Collected_By <chr>, Park <chr>, Scientific_Name <chr>,
# Functional_Type <chr>, Mature_Size <fct>, Native <chr>, Edible <chr>,
# Nuisance <chr>, Structural_Value <dbl>, Carbon_Storage_lb <dbl>,
```

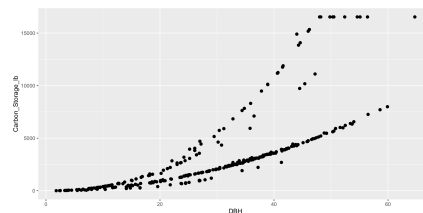
Q: What does a row represent here?

26

Scatterplots

- Explore relationships between numerical variables.
 - We will be especially interested in **linear** relationships.

```
1 ggplot(data = near_Reed,
2       mapping = aes(x = DBH,
3                     y = Carbon_Storage_lb)) +
4   geom_point(size = 2)
```

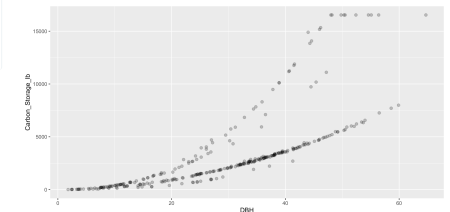


Is there something visually off with the points in this graph?

27

Scatterplots

```
1 ggplot(data = near_Reed,
2       mapping = aes(x = DBH,
3                     y = Carbon_Storage_lb)) +
4   geom_point(size = 2, alpha = 0.25)
```

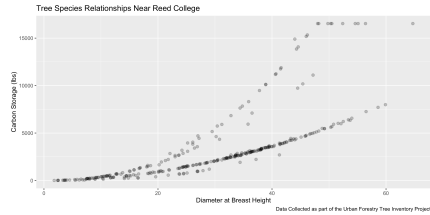


- Fix over-plotting (using **alpha**)
- What's going on in this graph?

28

Adding plot titles and axis labels

```
1 ggplot(data = near_Reed,
2       mapping = aes(x = DBH,
3                     y = Carbon_Storage_lb)) +
4   geom_point(size = 2, alpha = 0.25) +
5   labs(x = "Diameter at Breast Height",
6        y = "Carbon Storage (lbs)",
7        caption = "Data Collected as part of the Urban Fi",
8        title = "Tree Species Relationships Near Reed Co
```

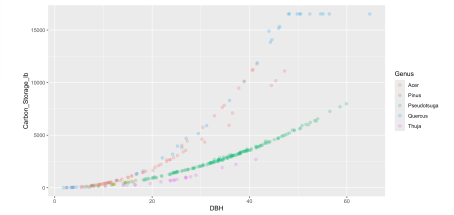


- Fix over-plotting (using `alpha`)
- Add a **plot title** and **axis labels** using `labs`
- What's going on in this graph? (labels help add context)

29

Scatterplots

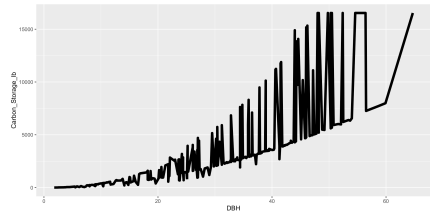
```
1 ggplot(data = near_Reed,
2       mapping = aes(x = DBH,
3                     y = Carbon_Storage_lb,
4                     color = Genus)) +
5   geom_point(size = 2, alpha = 0.25)
```



30

Linegraphs

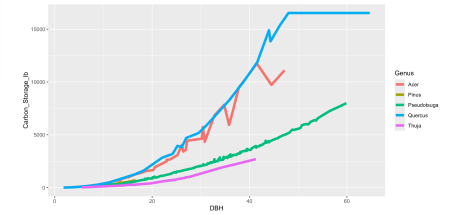
```
1 ggplot(data = near_Reed,
2       mapping = aes(x = DBH,
3                     y = Carbon_Storage_lb)) +
4   geom_line(linewidth = 2)
```



31

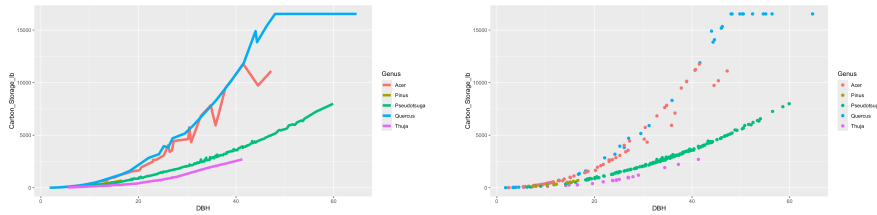
Linegraphs

```
1 ggplot(data = near_Reed,
2       mapping = aes(x = DBH,
3                     y = Carbon_Storage_lb,
4                     color = Genus)) +
5   geom_line(linewidth = 2)
```



32

Linegraphs vs scatterplots



- Similar, but one might be preferred depending on context.

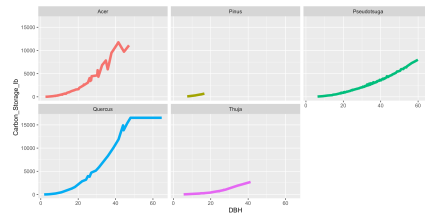
33

A speedy peek at advanced techniques!

34

Faceting

```
1 ggplot(data = near_Reed,
2       mapping = aes(x = DBH,
3                     y = Carbon_Storage_lb,
4                     color = Genus)) +
5   geom_line(linewidth = 2) +
6   facet_wrap(~Genus) +
7   guides(color = "none")
```



- Faceting is used to split one graphic into several smaller ones, based on the values of a categorical variable

35

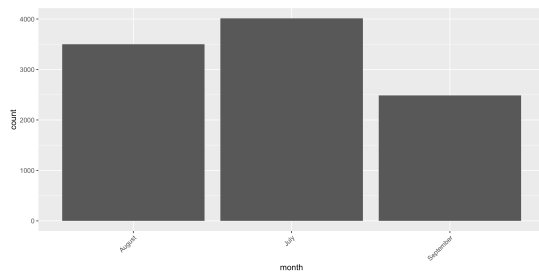
Customizing your ggplot2 Plots

- There are so **many** ways you can customize the look of your **ggplot2** plots
 - Adjusting labels
 - Zooming in
 - Using multiple **geoms**
 - Color!
 - Themes

36

Adjusting Labels: Rotate

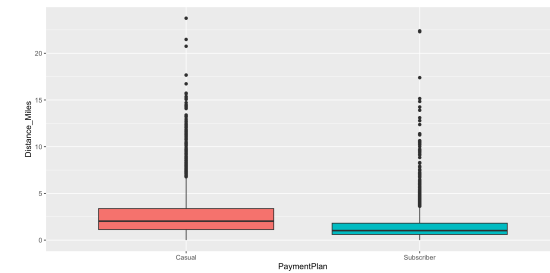
```
1 biketown <- biketown %>% mutate(month = case_when(month == "7" ~ "July",
2                                                    month == "8" ~ "August",
3                                                    month == "9" ~ "September"))
4 ggplot(data = biketown,
5       mapping = aes(x = month)) +
6   geom_bar() +
7   theme(axis.text.x =
8         element_text(angle = 45,
9                       vjust = 1,
10                      hjust = 1))
```



37

Zooming In

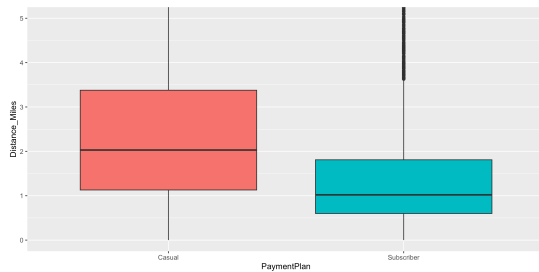
```
1 ggplot(data = biketown,
2       mapping = aes(x = PaymentPlan,
3                     y = Distance_Miles,
4                     fill = PaymentPlan)) +
5   geom_boxplot() +
6   guides(fill = "none")
```



38

Zooming In

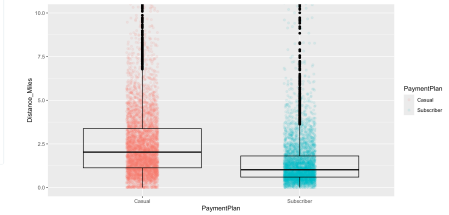
```
1 ggplot(data = biketown,
2       mapping = aes(x = PaymentPlan,
3                     y = Distance_Miles,
4                     fill = PaymentPlan)) +
5   geom_boxplot() +
6   guides(fill = "none") +
7   coord_cartesian(ylim = c(0, 5))
```



39

Multiple geoms

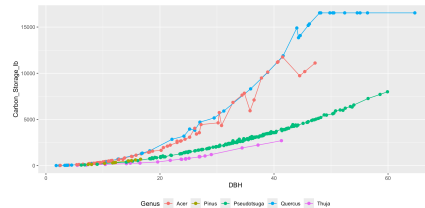
```
1 ggplot(data = biketown,
2       mapping = aes(x = PaymentPlan,
3                     y = Distance_Miles,
4                     color = PaymentPlan)) +
5   guides(fill = "none") +
6   coord_cartesian(ylim = c(0, 10)) +
7   geom_jitter(width = 0.1,
8               height = 0,
9               alpha = 0.1) +
10  geom_boxplot(fill = NA, color = "black")
```



40

Multiple geoms

```
1 ggplot(data = near_Reed,
2       mapping = aes(x = DBH,
3                     y = Carbon_Storage_lb,
4                     color = Genus)) +
5   geom_line() +
6   theme(legend.position = "bottom") +
7   geom_point(size = 2)
```



41

Change the Color

```
1 colors()
[1] "white"           "aliceblue"       "antiquewhite"
[4] "antiquewhite1"   "antiquewhite2"   "antiquewhite3"
[7] "antiquewhite4"   "aquamarine"      "aquamarine1"
[10] "aquamarine2"     "aquamarine3"     "aquamarine4"
[13] "azure"           "azure1"          "azure2"
[16] "azure3"          "azure4"          "beige"
[19] "bisque"          "bisque1"         "bisque2"
[22] "bisque3"         "bisque4"         "black"
[25] "blanchedalmond"  "blue"            "blue1"
[28] "blue2"           "blue3"           "blue4"
[31] "blueviolet"      "brown"           "brown1"
[34] "brown2"          "brown3"          "brown4"
[37] "burlywood"       "burlywood1"      "burlywood2"
[40] "burlywood3"      "burlywood4"      "cadetblue"
```

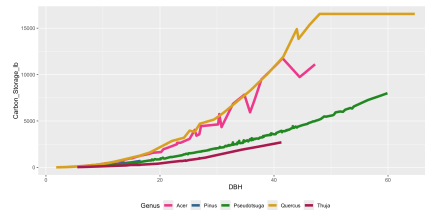
You can also use hex color codes to fully customize colors.

This linked webpage is a permitted internet resource for going to town on color customization

42

Change the Color

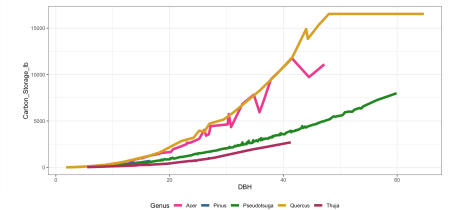
```
1 ggplot(data = near_Reed,
2       mapping = aes(x = DBH,
3                     y = Carbon_Storage_lb,
4                     color = Genus)) +
5   geom_line(size = 2) +
6   theme(legend.position = "bottom") +
7   scale_color_manual(values = c("violetred2",
8                                "steelblue4",
9                                "forestgreen",
10                               "goldenrod",
11                               "#aa0951"))
```



43

Use a Different Theme

```
1 ggplot(data = near_Reed,
2       mapping = aes(x = DBH,
3                     y = Carbon_Storage_lb,
4                     color = Genus)) +
5   geom_line(size = 2) +
6   scale_color_manual(values = c("violetred2",
7                                "steelblue4",
8                                "forestgreen",
9                                "goldenrod",
10                               "maroon")) +
11   theme_bw() + # Preset theme
12   theme(legend.position = "bottom") # Customize detail
```



44

- **Be warned:** Hours of my life have been lost to this, it can happen to you too...

Recap: ggplot2

```
1 library(tidyverse)
2 ggplot(data = ---, mapping = aes(---)) +
3   geom_---(---)
```

- This is the major concept to remember
 - Plug in **histogram**, **boxplot**, **bar**, **point**, or **line** for the **geom_---**
 - **Aesthetic mappings** required in **aes()** depend on the **geom** (see previous slides)
- Can then add more layers to add detail/context
- We covered a lot of ground - it takes **lots** of practice to become fluent in the details!

Next time:

- Data wrangling!